

Chapter 24: Eigenvalues and eigenvectors

Definition: Let A be an $n \times n$ matrix. If λ is a scalar and $\mathbf{v}$ is a nonzero vector in $\mathbb{R}^n$ such that

$$A\mathbf{v} =$$

then we say that λ is an **eigenvalue** of A , and $\mathbf{v}$ is an **eigenvector** of A associated with λ .

Example: Check that $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} 1 \\ 6 \end{bmatrix}$ are eigenvectors for $A = \begin{bmatrix} 3 & -1 \\ 6 & -4 \end{bmatrix}$, and find the corresponding eigenvalues.

Theorem: If $\mathbf{v}$ is an eigenvector of A , then $r\mathbf{v}$ ($r \neq 0$) is also an eigenvector of A .

Proof:

Definition: Let λ be an eigenvalue of an $n \times n$ matrix A , and define the **eigenspace** of A corresponding to λ to be

$$E_\lambda(A) = \{\mathbf{v} \in \mathbb{R}^n \mid A\mathbf{v} = \lambda\mathbf{v}\}$$

Theorem: $E_\lambda(A)$ is a subspace of $\mathbb{R}^n$.

Proof:

Finding eigenvalues and eigenvectors

Example: Let $A = \begin{bmatrix} 3 & 0 \\ -1 & -4 \end{bmatrix}$ be a matrix. Find the eigenvalues and eigenvectors of A .

More room for example:

Definition: Let

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$

be an $n \times n$ matrix. Then

$$\det(\lambda I_n - A) = \det \begin{bmatrix} \lambda - a_{11} & -a_{12} & \cdots & -a_{1n} \\ -a_{21} & \lambda - a_{22} & \cdots & -a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ -a_{n1} & -a_{n2} & \cdots & \lambda - a_{nn} \end{bmatrix}$$

is called the **characteristic polynomial** of A , denoted $c_A(\lambda)$ in your book.

The equation

$$\det(\lambda I_n - A) = 0$$

is called the **characteristic equation** of A .

Theorem: Let A be an $n \times n$ matrix. The eigenvalues of A are the roots of the characteristic polynomial of A .

Example: Find the characteristic polynomial of A if $A = \begin{bmatrix} 2 & 3 & 0 \\ 3 & 2 & 0 \\ -2 & -4 & 4 \end{bmatrix}$, then find the eigenvalues of A .

Finding eigenvalues and associated eigenvectors of matrix A

1. Determine the roots of the characteristic polynomial

$$c_A(\lambda) = \det(\lambda I_n - A).$$

These are the *eigenvalues* of A .

2. For each eigenvalue λ , find **all** nontrivial solutions to the homogeneous system

$$(\lambda I_n - A)\mathbf{x} = \mathbf{0}.$$

These are the *eigenvectors* of A associated with λ .

Warning: Do not make the common mistake of first transforming A to RREF and then finding the eigenvalues and eigenvectors of RREF.

Example: Find the eigenvalues and a basis for each eigenspace of $A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$.

Chapter 25 & 26: Diagonalization and Similarity

Definition: If A and B are $n \times n$ matrices, we say A and B are **similar**, and write $A \sim B$ if there is an invertible matrix P with $B = P^{-1}AP$.

Fact: If $A \sim B$, then $B \sim A$.

Proof:

Properties shared by similar matrices

Definition: The **trace** $\text{tr}(A)$ of an $n \times n$ matrix A is the sum of the entries on the main diagonal.

Example: The trace of $A = \begin{bmatrix} -2 & 2 & -4 \\ -7 & 3 & 6 \\ 4 & 1 & 0 \end{bmatrix}$ is $\text{tr}(A) =$

Fact: For two $n \times n$ matrices A and B and scalar c ,

- a. $\text{tr}(A + B) = \text{tr}(A) + \text{tr}(B)$.
- b. $\text{tr}(cA) = c \text{tr}(A)$
- c. $\text{tr}(AB) = \text{tr}(BA)$ (However, $\text{tr}(AB)$ does NOT necessarily equal $\text{tr}(A) \text{tr}(B)$).

Proof of (c) (if time):

Theorem: If A and B are similar matrices, then A and B have the same determinant, rank, trace, characteristic polynomial, and eigenvalues.

Proof of Trace:

Proof of Characteristic Polynomial/Eigenvalues:

Warning:

- Sharing all five properties does NOT guarantee two matrices are similar.
- Being similar is NOT the same as being row equivalent.

Definition: An $n \times n$ matrix A is called **diagonalizable** if $P^{-1}AP$ is a diagonal matrix for some invertible matrix P .

In other words: A is **diagonalizable** if A is *similar* to a diagonal matrix.

Example: Suppose $A = \begin{bmatrix} 3 & 1 \\ -2 & 0 \end{bmatrix}$. Compute $P^{-1}AP$ using $P = \begin{bmatrix} -1/2 & -1 \\ 1 & 1 \end{bmatrix}$ (note that $P^{-1} = \begin{bmatrix} 2 & 2 \\ -2 & -1 \end{bmatrix}$). What does this tell you about A ?

What are the eigenvalues of A ?

What is A^5 ?

So diagonalizing a matrix is *nice*. How do we do it?

Suppose $P^{-1}AP = D$ for some diagonal matrix $D = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$ for some unknown λ_i s. Write $P = [\mathbf{v}_1 \mathbf{v}_2 \cdots \mathbf{v}_n]$. Note that $P^{-1}AP = D$ if and only if $AP = PD$.

What happens when I take PD ?

What happens when I take AP ?

Theorem: An $n \times n$ matrix A is diagonalizable if and only if A has n eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ such that the matrix $P = [\mathbf{v}_1 \mathbf{v}_2 \cdots \mathbf{v}_n]$ is invertible. In this case, $P^{-1}AP = D$, where D is a diagonal matrix such that the elements of D on the main diagonal are the eigenvalues of A .

Said another way:

Theorem: Let A be an $n \times n$ matrix. A is diagonalizable if and only if $\mathbb{R}^n$ has a basis $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ consisting of eigenvectors of A .

Example: Let $A = \begin{bmatrix} 3 & 0 \\ -1 & -4 \end{bmatrix}$. Is A diagonalizable?

Theorem: Let $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k$ be eigenvectors corresponding to distinct eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_k$ of an $n \times n$ matrix A . Then $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ is a linearly independent set.

Proof:

Theorem: If A is an $n \times n$ matrix with n distinct eigenvalues, then A is diagonalizable.

Proof:

Example: Let $A = \begin{bmatrix} 4 & 1 \\ 0 & 4 \end{bmatrix}$. Is A diagonalizable?

Does an $n \times n$ matrix need to have n distinct eigenvalues to be diagonalizable?

If the roots of the characteristic polynomial of A are not all distinct, then A may or may not be diagonalizable.

The characteristic polynomial of an $n \times n$ matrix is a degree n polynomial. It always has n **complex** (not necessarily distinct) **roots**, and can be written as the product of n factors:

$$p(\lambda) = (\lambda - \lambda_1)^{k_1} (\lambda - \lambda_2)^{k_2} \dots (\lambda - \lambda_r)^{k_r},$$

where $\lambda_1, \lambda_2, \dots, \lambda_r$ are distinct eigenvalues of A , and $k_1, k_2, \dots, k_r$ are positive integers whose sum is n .

Definition: If A is an $n \times n$ matrix whose characteristic polynomial p is

$$p(\lambda) = (\lambda - \lambda_1)^{k_1} (\lambda - \lambda_2)^{k_2} \dots (\lambda - \lambda_r)^{k_r},$$

then k_i is called the **algebraic multiplicity** of λ_i .

Example: What are the eigenvalues of A and the algebraic multiplicity of each eigenvalue if

$$A = \begin{bmatrix} 2 & 4 & -1 & 3 & 7 \\ 0 & -5 & 2 & 2 & 0 \\ 0 & 0 & 2 & 8 & -3 \\ 0 & 0 & 0 & 3 & 9 \\ 0 & 0 & 0 & 0 & -5 \end{bmatrix} ?$$

Definition: If λ is an eigenvalue of an $n \times n$ matrix A , the **eigenspace** of A corresponding to λ is

$$E_\lambda(A) = \{\mathbf{x} \in \mathbb{R}^n \mid A\mathbf{x} = \lambda\mathbf{x}\}$$

We saw earlier that $E_\lambda(A)$ is a subspace of $\mathbb{R}^n$, so it must have a dimension. In fact,

$$\dim E_\lambda(A) = \dim \text{null}(\lambda I_n - A)$$

$\dim E_\lambda(A)$ is called the **geometric multiplicity** of λ .

Fact: Let λ be an eigenvalue of algebraic multiplicity m of a square matrix A . Then $\dim E_\lambda(A) \leq m$, i.e.

Theorem: A matrix A is diagonalizable if and only if each eigenvalue has geometric multiplicity equal to its algebraic multiplicity.

Example: Find the algebraic and geometric multiplicity of each eigenvalue of

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 2 & 3 \\ 0 & 0 & 3 \end{bmatrix}.$$

Note that, if A is similar to a diagonal matrix D , and the characteristic polynomial of A has *non-real* roots, then the diagonal of D will have *non-real* numbers!

So A is diagonalizable, but not as a *real* diagonal matrix.

Example: Let $A = \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}$ be a 2×2 matrix. Is A diagonalizable?

Fact: A symmetric $n \times n$ matrix will have ONLY real eigenvalues.

(Extra examples if time):

Example: Is $A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ diagonalizable?

Example: Is $B = \begin{bmatrix} -1 & -3 & 0 \\ 6 & 8 & 0 \\ 3 & 3 & 2 \end{bmatrix}$ diagonalizable?